

Fig. 4: The transmitted waves after being received by the target are reradiated in all directions, with some of them going towards the radar antenna



Fig. 5: Program screenshot when the timer goes 'off'

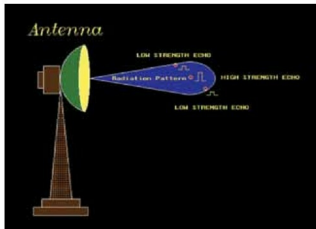


Fig. 6: Maximum power is in the direction of the axis of the beam and it diminishes rapidly in directions away from the axis

target (refer Fig. 2). First, signal waves from the radar transmitter are transmitted to the duplexer. The duplexer acts as a transmitter and transmits the signals to space through the antenna. The signal echoes reflected off the target are received by the same antenna and

sent to the duplexer.

Display. The display provides the operator a continuous, easily understandable graphical picture of the relative position of the objects.

Second part. This part demonstrates the process of distance determi-

nation. In the program, an antenna, a target and a timer are shown. As soon as the antenna starts radiating the RF waves towards the target, the timer turns 'on' (refer Fig. 3). These waves after being received by the target are reradiated in all directions, with some of them directed towards the radar antenna (refer Fig. 4).

When the waves are received by the antenna, the timer switches to its 'off' state (refer Fig. 5). The 'on' time duration of the timer is then used to determine the distance of the target from the radar station.

Transmitter. The radar transmitter produces short-duration high-power RF pulses, which are sent into space through the antenna.

Duplexer. The duplexer alternately switches the antenna between the transmitter and receiver, so only one antenna is required. This switching is necessary. Otherwise, high-power RF pulses, if allowed to enter the receiver, would destroy it.

Radar antenna. When the signal is transmitted into the space, the reflected signal beam is received by the receiver through the antenna.

Receiver. The receiver amplifies and demodulates RF signals and provides

them to the display.

Display. The display provides the operator a continuous, easily understandable graphical picture of the relative position of the objects.

Third part. This part demonstrates determination of the target's direction with respect to the radar. It shows the radiation pattern and the strength of the signal at three different positions. It depicts relative power in the same plane existing at the same distances from the antenna or from the origin of the radar beam. Maximum power is in the direction of the axis of the beam and it diminishes rapidly in directions away from the axis (refer Fig. 6). This signal strength is then used and, with the help of servo mechanism, the antenna gets directed towards the target.

Program functions

Initgraph(&gd,&gm,"c:\\turbo3\\bgi"). Initialises the graphics system by loading a graphics driver from the disk and then putting the system into graphics mode. It also resets all graphics settings like colour, palette and current position to their defaults.

***gd.** Specifies the graphics driver to be used.

***gm.** Specifies the initial graphics mode (unless *gd=DETECT). When *gd=DETECT, initgraph sets the graphics mode to the highest resolution available for the detected driver.

Closegraph(). Switches back the screen from graphics mode to text mode. It is called after getch() function so that the screen doesn't clear until the user hits the key.

Void principle(). Demonstrates the process of wave-flow from the transmitter to the target and display of the